Online Analytical Processing (OLAP)

Multidimensional data analysis and its four core operations



OLAP organizes warehouse data into a **cube**: dimensions such as time, product and location, with a measure (sales, profit) in every cell. Analysts view the same data from many angles and answer complex business questions in seconds.

Types: MOLAP, ROLAP and HOLAP balance speed and storage differently.

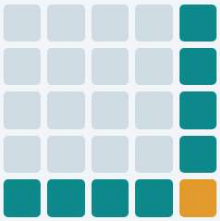
Used for: reporting, forecasting, budgeting and trend analysis.

Presenting OLAP and its operations

Roll-up

Summarize up a hierarchy

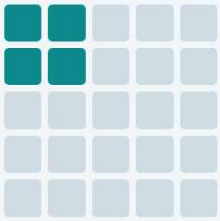
Aggregates detail into totals.
City → State → Country



Drill-down

Move from summary to detail

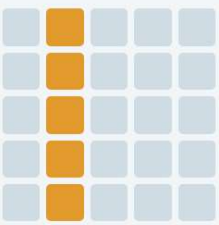
Splits totals into finer levels.
Year → Quarter → Month



Slice

Fix one dimension

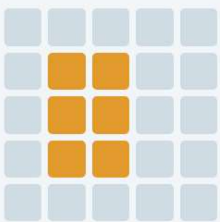
Gives a smaller sub-cube.
Time = Q1, all products and regions



Dice

Select on two or more dimensions

Extracts a focused sub-cube.
Q1–Q2, Phone or Laptop, Asia



Topic: OLAP Operations **Name:** _____ **Class:** _____